Code:

```
abstract class EnergySource {
    String sourceId;
    String sourceName;
    double energyGenerated;

    EnergySource(String sourceId, String sourceName, double
energyGenerated) {
        this.sourceId = sourceId;
        this.sourceName = sourceName;
        this.energyGenerated = energyGenerated;
    }


    abstract double calculateEfficiency();


    void displayDetails() {
        System.out.println("Source ID: " + sourceId);
        System.out.println("Source Name: " + sourceName);
        System.out.println("Energy Generated: " +
energyGenerated + " kWh");
        System.out.printf("Efficiency: %.2f\n",
calculateEfficiency());
        System.out.println("----------------------------------
-");
    }
}


class SolarEnergy extends EnergySource {

    SolarEnergy(String sourceId, String sourceName, double
energyGenerated) {
        super(sourceId, sourceName, energyGenerated);
    }


    double calculateEfficiency() {
        return (energyGenerated / 5000.0) * 100.0;
    }
}


class WindEnergy extends EnergySource {
```

```
    WindEnergy(String sourceId, String sourceName, double
energyGenerated) {
        super(sourceId, sourceName, energyGenerated);
    }

    double calculateEfficiency() {
        return (energyGenerated / 8000.0) * 100.0;
    }
}


public class EnergyManagementSystem {
    public static void main(String[] args) {

        EnergySource ref;

        ref = new SolarEnergy("SOL-11", "Solar Park", 2500.0);
        ref.displayDetails();


        ref = new WindEnergy("WND-22", " Energy Center",
7000.0);
        ref.displayDetails();

        System.out.println("Mahammad Salman Sameer");
        System.out.println("1BF25CS143");
    }

}
```

Output:

```
 ----jGRASP exec: java EnergyManagementSystem
Source ID: SOL-11
Source Name: Solar Park
Energy Generated: 2500.0 kWh
Efficiency: 50.00
---------------------------------
Source ID: WND-22
Source Name:  Energy Center
Energy Generated: 7000.0 kWh
Efficiency: 87.50
---------------------------------
Mahammad Salman Sameer
1BF25CS143

 ----jGRASP: Operation complete.
```